

Vasopressor and Drug Management of Intraoperative Hypotension

2022 ESA/EBA Guidelines on Haemodynamic Monitoring and Management

1. First-Line Vasopressors

PHENYLEPHRINE (alpha-1 agonist):

- Mechanism: Pure vasoconstriction; increases SVR without chronotropy
- Dose: IV bolus 50–200 mcg; infusion 25–100 mcg/min
- Indication: Vasodilatory IOH (spinal, volatile agents); normal cardiac function
- Caution: Reflex bradycardia may reduce cardiac output; avoid in low CO states
- Onset: <1 minute; Duration: 5–20 minutes

EPHEDRINE (mixed alpha/beta agonist):

- Mechanism: Increases HR, contractility, and SVR
- Dose: IV bolus 5–10 mg; repeat every 3–5 min as needed
- Indication: IOH with bradycardia; spinal anesthesia-induced hypotension
- Caution: Tachyphylaxis occurs rapidly; associated with fetal acidosis (obstetric use)
- Note: Crosses the blood-brain barrier; can cause agitation or awareness

2. Second-Line & Rescue Vasopressors

NOREPINEPHRINE (alpha-1 > beta-1 agonist):

- Mechanism: Potent vasoconstriction + moderate inotropy
- Dose: Infusion 0.01–0.3 mcg/kg/min
- Indication: Refractory IOH; septic shock intraoperatively; high-risk cardiac patients
- Central line preferred for sustained infusion

VASOPRESSIN (V1 receptor agonist):

- Mechanism: Vasoconstriction independent of adrenergic receptors
- Dose: 0.03–0.04 units/min infusion
- Indication: Catecholamine-resistant vasodilatory shock; adjunct to norepinephrine

- Advantage: Preserved efficacy during acidosis where catecholamines lose potency

DOPAMINE (dose-dependent):

- Low dose (1–3 mcg/kg/min): dopaminergic, renal vasodilation
- Moderate dose (3–10 mcg/kg/min): beta-1 dominant, increases CO
- High dose (>10 mcg/kg/min): alpha-1 dominant, increases SVR
- Generally second-line; higher arrhythmia risk than norepinephrine

3. Preventive Pharmacology

PRE-EMPTIVE PHENYLEPHRINE INFUSION (for spinal anesthesia):

- Starting a phenylephrine infusion at 25–50 mcg/min at induction of spinal block prevents >50% of spinal-induced IOH episodes (Cochrane meta-analysis, 2020).

METHOXAMINE: Long-acting alpha-1 agonist used in some centers for sustained prevention; not widely available in US/EU; 15–20 minute duration.

CO-PHENYLCAINE: Not used systemically; topical use only.

ATROPINE: Not a vasopressor but addresses bradycardia-mediated IOH;
IV 0.5–1 mg bolus; may be combined with phenylephrine for mixed pictures.

EPINEPHRINE: Reserved for anaphylaxis-induced IOH or cardiac arrest;
0.1 mg IV bolus; titrate carefully due to extreme potency.